WASP: function from structural neighbourhoods

WASP: a pipeline for functional annotation prediction based on AlphaFold structural models

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Journal Club

The question: what can structure rescue?

Sequence divergence can obscure related proteins even when their folds remain conserved.

1. Can structural search recover functional labels missed by sequence search?
2. Does a network of related structures improve on copying one best hit?
3. Can those predictions identify native enzymes for orphan metabolic reactions?

Central contribution: a proteome-scale workflow combining Foldseek, reciprocal hits and local functional enrichment.

Introduction; Fig. 1. Predictions are hypotheses about function, not biochemical validation.

A motivating example: FtsZ and tubulin

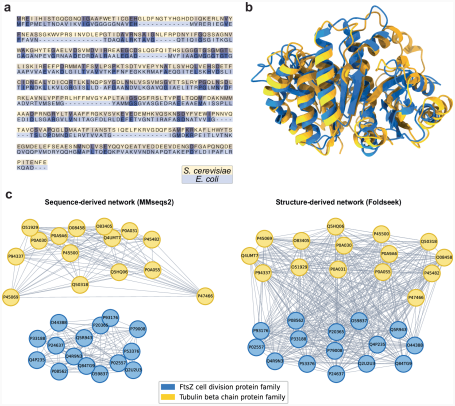


Fig. 2a–c, reproduced; Supplementary Fig. 1. CC BY 4.0.

E. coli FtsZ versus yeast tubulin

- About 15% sequence identity.
- Structural TM-score: 0.60 (1 = perfect match).
- In a sample of 15 family members, sequence search separates groups; structural search connects them.

This motivates remote family recognition; an equivalent fold alone does not establish identical cellular roles.

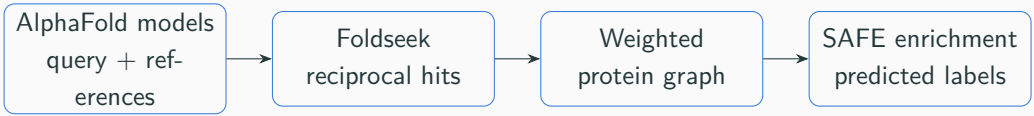
Inputs and outputs before the algorithm

Input	Organism NCBI taxonomy ID; AlphaFold protein models. A new proteome requires structure prediction first.
References	SwissProt plus AFDB clustered at 50% sequence identity and 90% bidirectional coverage; query proteome included for paralogues.
Outputs	Structural networks, predicted labels and enrichment scores; collateral predictions for reference proteins.
Labels	Pfam, CATH, PANTHER, GO, EC numbers and Rhea reaction IDs.

Study snapshot: AFDB v4, accessed September 2024. All compared structures are predicted models, including when a PDB structure exists.

Methods: WASP implementation details; Data availability.

Search, reciprocate, connect, enrich



1. Foldseek encodes local tertiary interactions with its 3Di alphabet for fast structural search.
2. Keep significant bidirectional matches within top-hit neighbourhoods.
3. Infer labels from enrichment in a node's local graph neighbourhood.

Fig. 1; Results: network representation; Methods. Schematic redrawn.

Reciprocal hits reduce one-sided matches

For query q and reference r, retain a structural relationship when both searches support it within the selected neighbourhood.

$$q \in \mathrm{TopK}(r), \qquad r \in \mathrm{TopK}(q)$$

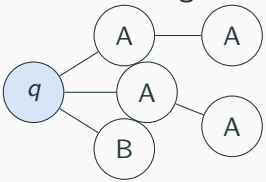
- Default alignment filters: bitscore > 50 and E-value $< 10^{-10}$.
- Start with $K = 10$; increase by 10 at each iteration; default three iterations.
- Protein nodes; undirected edges weighted by normalised alignment bitscores.

Trade-off: expanding the neighbourhood can reconnect isolated proteins, but also admit less specific functional relationships.

Methods: structural alignment and functional inference. Top-K is a teaching abstraction of the reciprocal neighbourhood rule.

How enrichment differs from copying a best hit

Illustrative local neighbourhood



A and B denote known functional labels.

SAFE tests whether a label is overrepresented near a protein compared with its background frequency.

- Neighbours may be more than one edge away.
- One-tailed hypergeometric/Fisher test; multiple-testing correction.
- Significant labels ($p < 0.05$ after correction) receive log-transformed enrichment scores.

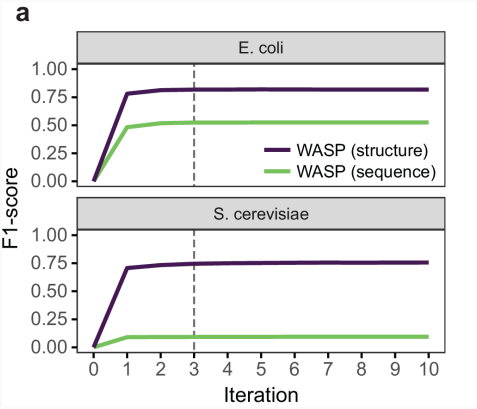
Methods: functional inference. Graph is illustrative, not paper data. Enrichment is not a calibrated probability of enzyme activity.

What the benchmarks actually test

Test	Design and interpretation
Hidden labels	1,142 E. coli and 1,108 yeast proteins with all six descriptors; mask query labels, recover from other network members.
Sequence ablation	Replace Foldseek by MMseqs2, keep network/enrichment workflow. Tests search representation within this pipeline.
Enzyme benchmarks	New-392: 392 enzymes, 177 EC labels; SwissProt data after CLEAN training. Price-149: experimentally characterised enzymes.
Comparators	EggNOG-mapper, CLEAN, DeepEC; ProteInfer scores from CLEAN study; best-hit structural label transfer discussed in text.

Methods: WASP benchmark. Weighted multilabel precision/recall/F1; comparative enzyme runs use ten iterations.

Structural search preserves many more query proteins



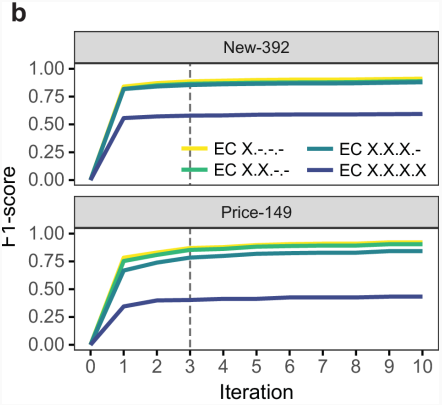
The ablation isolates a coverage bottleneck.

- MMseqs2 variant excludes about 43% of E. coli and 90% of yeast proteins for lack of reciprocal hits.
- Structural WASP plateaus early, motivating three default iterations.
- Figure shows roughly 0.8 E. coli and 0.75 yeast F1 at plateau.

Caution: selected, well-annotated proteins are easier than completely unknown families.

Fig. 3a, cropped, CC BY 4.0; Supplementary Figs. 2–4. Figure values shown approximately; text's "above 80%" is broader.

The fourth EC level is substantially harder



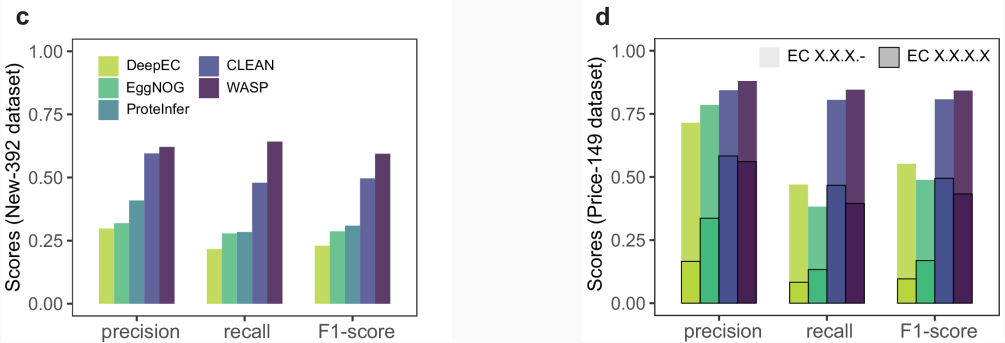
EC levels move from broad reaction classes to more specific enzyme assignments.

- Three-level EC F1 exceeds about 0.85 after ten iterations.
- Full four-level EC F1: about 0.60 on New-392; about 0.40 on Price-149.
- Strong fold/family recognition does not guarantee exact substrate specificity.

Always report the EC resolution when comparing tools.

Fig. 3b, cropped, CC BY 4.0; Results: hidden-label benchmarks.

Comparative accuracy depends on dataset and label resolution



New-392: WASP leads plotted sequence-based tools. Price-149: similar to CLEAN for full EC labels; stronger at three EC levels. Text reports best-hit transfer wins New-392 but degrades on Price-149; that comparator is not displayed in these panels.

Fig. 3c–d, cropped, CC BY 4.0. Outlined bars in d: full EC; unoutlined: three levels. Supplementary Figs. 5–6: complementarity with CLEAN.

Proteome annotation: useful coverage, incomplete specificity

1. **20 industrially relevant organisms:** up to 20–30% of previously uncharacterised proteins receive predicted annotations. Coverage varies by organism and descriptor.
2. **New Yarrowia lipolytica strain:** 6,575 of 6,577 ORFs modelled; the two $> 4,500$-residue proteins exceed hardware limits.
3. **New-strain coverage:** about 30% for family/domain/GO descriptors versus about 10% for EC/Rhea. EggNOG has the highest overall coverage; WASP adds about 200 uniquely annotated proteins.

An extra label is added knowledge only if the transferred function is correct.

Table 2; Results: industrial organisms and full genome prediction; Supplementary Figs. 7–12.

Reverse annotation: find the gene for an orphan reaction

1. Extract GEM reactions without a gene association. Exclude exchange, spontaneous, biomass and other non-enzymatic reactions.
2. Map each reaction to EC/Rhea; obtain experimentally supported SwissProt reference enzymes with AlphaFold models.
3. Group references by catalytic function and structural subclusters (TM-score > 0.5).
4. Search the native proteome; retain significant hits shared across all references in each subcluster; rank by average TM-score.

Output: candidate native enzymes to curate gene–protein–reaction rules. This is a specialised adaptation, with manual GEM preprocessing.

Methods: genome-scale model gap-filling adaptation; Fig. 4. No claim of new experimental enzyme confirmation.

The 75–100% recovery claim has a restricted denominator

Reported orphan-reaction accounting

All orphan reactions	7,441
With reference protein	1,900
Without reference	5,541

Only about **26%** of all orphans have a reference and enter the structural mapping analysis.

Among eligible reactions: at least one candidate for 75–100% across the 20 organisms; average hit TM-scores often > 0.7. **Main missing category:**

transporters. Of 3,529 EC-7 orphans, 3,369 lack a reference. Candidate recovery is not

recovery of all GEM gaps, nor proof of native catalysis.

Fig. 4a–b; Supplementary Table 2. Percent eligible calculated as 1,900/7,441.

A concrete candidate: yeast P38256

Problem: an orphan 4-hydroxybenzoyl-CoA thioesterase reaction (EC 3.1.2.23).

$$\text{4-hydroxybenzoyl-CoA} \longrightarrow \text{4-hydroxybenzoate}$$

- P38256 is ambiguously annotated as a probable thioesterase or acyltransferase.
- WASP links it to annotated thioesterases Q04416 and P56653.
- Structural similarity is high (TM-score about 0.72), despite only 12% sequence identity to P56653.

Interpretation: prioritise a testable gene–reaction hypothesis. An enzyme assay, substrate panel and genetic complementation would test specificity.

Fig. 4c; Results: mapping orphan reactions. Proposed validation is journal-club interpretation.

GEM refinement improves agreement with essentiality screens

Organism	Precision before	After
Helicobacter pylori	0.56	0.59
Pseudomonas aeruginosa	0.49	0.58
Mycobacterium tuberculosis	0.82	0.92

M. tuberculosis F1 rises from 0.77 to 0.81. Models use media matched as closely as possible to transposon screens.

$$(g_1 \text{ AND } g_2) \quad \rightarrow \quad ((g_1 \text{ OR } g_3) \text{ AND } g_2)$$

Adding candidate redundancy can fix false essentiality predictions. But false positives were targeted using the screen itself: this is curation agreement, not an untouched prospective test.

Results and Methods: integration into GEMs. Precision = TP/(TP+FP). GPR example adapted from Methods.

What remains uncertain?

1. **Annotation dependence:** errors can propagate; clusters with no known labels remain dark. Homologue masking is not family-level or temporal holdout.
2. **Specificity:** global alignment does not emphasise catalytic residues. Similar folds can catalyse different reactions; the full-EC performance exposes this limit.
3. **Biological validation:** gene–reaction mappings and drug-target proposals remain candidates. Rv2172c docking (Supplementary Fig. 14) does not establish binding or inhibition.

Practical conclusion: use WASP alongside sequence-based methods to prioritise experiments, retaining provenance and confidence for each transferred label.

Discussion; Fig. 3; Supplementary Fig. 14. Benchmark leakage/validation qualifications are journal-club assessment.

Discussion and reproducibility

1. Would WASP retain its advantage after excluding close sequence relatives and using a common historical annotation snapshot?
2. Which experimental assay would most efficiently discriminate the top candidate enzyme from a fold-similar paralogue?
3. Should GEM refinement be judged on independent conditions or a second transposon dataset after curation?

Paper: doi:10.1038/s41467-026-75956-z
Code: github.com/gioodm/wasp-proteins-annotation
Archived production version: doi:10.5281/zenodo.20617135
Dependencies include Foldseek and gsutil; default run: three enrichment iterations.

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